

ARYAN RAJPUT

Vadodara, Gujarat

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LinkedIn

GitHub

Medium

Objective

Computer Science student focused on backend development and machine learning, with interests in scalable systems, software engineering, and applied AI.

Experience

Horizontal Integration India Pvt Ltd

Jan 2026 – May 2026

SDE Intern / DX Intern

Vadodara, Gujarat

- Developed modular frontend system for an enterprise training portal using React, TypeScript, Tailwind CSS, and React Router, contributing across 10+ routed views and improving UI scalability through reusable workflows.
- Implemented Atomic Design principles to build 25+ reusable React components, reducing UI duplication by ~30% and improving maintainability across shared frontend modules.
- Built data-driven UI workflows using JSON-based rendering and component registry patterns, enabling dynamic generation of navigation systems, category listings, and training content with reduced hardcoded logic. Improved frontend reliability and debugging efficiency by leveraging structured state handling and TypeScript-driven validation, reducing development turnaround time by ~20%.

Projects

Gradcore | Python, NumPy | GitHub

- Built a scalar automatic differentiation engine and mini neural net library.
- Gained a low-level, intuitive understanding of how autograd engines and training loops work under the hood.
- Implemented core neural network components manually using only Python, with no external ML libraries.

ML Research Reproduction Projects | PyTorch, NumPy, Deep Learning | GitHub

- Recreated foundational neural network architectures from influential research papers to study architectural design choices and optimization behavior at a systems level.
- Implemented the Transformer architecture proposed in Attention Is All You Need, including positional encodings, multi-head attention, masking, and encoder-decoder attention mechanisms.
- Developed practical intuition for sequence modeling, gradient flow, tensor manipulation, and attention-based representation learning through from-scratch implementation and experimentation.

AddStack | Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT | GitHub

- Built a RESTful backend with 15+ API endpoints for user authentication, post management, voting, and content retrieval using FastAPI and PostgreSQL.
- Designed a normalized PostgreSQL schema with SQLAlchemy ORM and Alembic migrations, ensuring data integrity and maintainable database evolution.
- Implemented JWT-based authentication, authorization, password hashing, and request validation with Pydantic to secure and streamline API operations.

Technical Skills

Languages: HTML/CSS, JavaScript, TypeScript, Python, SQL

Web Libraries/Frameworks: Bootstrap, TailwindCSS, React, DOM Manipulation, FastAPI

Database/ORMs: PostgreSQL, SQLAlchemy, Sqlite3

Libraries & Utilities: NumPy, Pandas, Matplotlib, Pydantic

ML Technologies & Frameworks: Scikit-learn, PyTorch, Hugging Face Transformers

Tools & Technologies: Git/GitHub, Browser Devtools, bash, npm, uv, LaTeX

Education

Sardar Vallabhbhai Patel Institute of Technology

Aug 2022 – July 2026

B.E. in Computer Science and Design (CGPA: 8.19/10)

Gujarat Technological University, GJ

Relevant Coursework: Engg. Mathematics (Linear algebra, Probability and Stats, Calculus), ML, AI, Data Structures, Object Oriented Programming, Web Development, Cryptography and Network Security, DBMS, OS

Certifications and Volunteering

Short Term Training Programme (STTP) @ SVNIT Surat [Certificate]

July 2025

Recent Trends and Practice in Artificial Intelligence

Surat, GJ